

JAYANTH DASAMANTHARAO

New Brunswick, NJ — (848)313-5617 — jayanthdasamantharao@gmail.com — [linkedin.com/in/djayanth](https://www.linkedin.com/in/djayanth)

PROFESSIONAL EXPERIENCE

Senior AI Engineer, EliteUS Software Solutions

Aug 2024 - Present | Edison, NJ

- Researched and optimized deep learning perception models (CNNs, Transformers) for real-time deployment, applying quantization (static/dynamic), pruning, and hardware-aware optimization strategies.
- Built deployment pipelines exporting PyTorch models to ONNX and benchmarking latency, throughput, and memory across heterogeneous compute environments.
- Designed low-latency inference services (Python/C++), leveraging mixed precision (FP32/FP16/INT8) and systematic precision trade-off analysis to validate deterministic performance under real-time constraints.

Data Science Intern, Harvest Software Solutions

June 2023 - Sept 2023 | Jacksonville, Florida

- Developed and optimized CNN-based and transformer-based NLP models, applying model pruning and dynamic quantization techniques for efficient inference.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50% and developed evaluation frameworks to analyze performance trade-offs across precision formats (FP32, FP16, INT8).
- Integrated PyTorch models into production REST APIs, implementing performance monitoring, automated regression tracking to detect model drift and accuracy degradation across deployments.

Database Analyst, Accenture

June 2021 - Aug 2022 | Hyderabad, India

- Developed Python-based ETL and data processing pipelines for high-performance distributed systems, reducing compute overhead by 25% and transformed large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation for large-scale data aggregation across distributed warehouse systems (AWS Athena, Redshift).
- Applied algorithmic and memory optimization techniques for scalable processing of millions of records.

Researcher, Andhra University

January 2021 - May 2021 | Visakhapatnam, India

ADAS-Oriented Autonomous Driving Perception Research

- Designed and optimized perception models (CNNs, YOLO, Faster R-CNN, transformer-based detectors) for autonomous driving scenarios under ADAS-style real-time latency constraints.
- Developed a multi-sensor fusion pipeline integrating camera, LiDAR, and radar streams with synchronization and feature-level fusion for robust perception in dynamic environments.
- Applied hardware-aware optimizations (static/dynamic quantization, pruning, mixed precision), exported models to ONNX, benchmarked latency, throughput, memory across heterogeneous compute targets using a Unity3D simulation.

TECHNICAL SKILLS

Programming:	Python, C++, SQL. Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, RestAPI.
Machine Learning :	Model Fine-tuning, evaluation and experimentation, deployment, Feature Engineering.
Model Optimization:	Quantization, model pruning, distillation, operator fusion, memory-efficient architectures.
Model Deployment:	ONNX export, hardware-aware inference optimization, heterogeneous compute benchmarking.
Computer Vision:	Object detection, segmentation, CNN architectures, transformer-based perception models.
Accelerator Exposure:	CUDA, GPU memory optimization, mixed precision (FP16/INT8), profiling, performance tuning.
Cloud & MLOps:	PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science

Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering

2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Hardware-Aware Perception Model Optimization: Researched quantization and compression strategies for CNN-based perception models; implemented ONNX-based export workflows and benchmarked performance across precision formats to achieve improved latency and energy efficiency.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>